



# A Data-Driven Competitive Brand Perception Framework for Strategic Brand Positioning Through Social Media Analytics

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**Abstract:** Brand perception has increasingly been recognized as a critical source of competitive intelligence in digital markets, yet existing studies have largely examined consumer sentiment, discussion topics, or product attributes independently, thereby limiting their ability to support strategic brand positioning. A competitive brand perception framework was proposed to address this limitation by integrating sentiment analysis, latent Dirichlet allocation-based topic modeling, and aspect-based sentiment analysis into a unified analytical framework capable of transforming consumer-generated social media data into competitive intelligence. The framework was validated using 12,603 posts collected from X (formerly Twitter) concerning four leading beauty and personal care brands—Dove, Nivea, Estée Lauder, and L'Oréal. Overall consumer attitudes were quantified through sentiment analysis, dominant perception themes were identified using latent Dirichlet allocation-based topic modeling, and evaluations of brand- and product-related attributes were examined through aspect-based sentiment analysis. Three complementary competitive positioning indicators—topic salience, aspect sentiment, and brand distinctiveness—were introduced. The findings demonstrate that overall sentiment alone is insufficient for explaining competitive positioning. Although Nivea generated the greatest volume of consumer discussion, the highest sentiment score and strongest topic distinctiveness were observed for Estée Lauder. Dove was primarily associated with cleansing and body care, whereas L'Oréal demonstrated a distinctive perceptual position centered on facial skincare and makeup-related discussions. These findings further indicate that competitive differentiation is shaped not only by the polarity of consumer evaluations but also by the thematic structure and attribute-specific associations embedded within online conversations. Consequently, brand perception should be conceptualized as a multidimensional and competitive construct rather than as a single sentiment-based metric. The proposed competitive brand perception framework extends the methodological foundation of social media analytics by integrating complementary analytical perspectives into a unified competitive intelligence framework and provides a practical decision-support tool for identifying perceptual advantages, uncovering differentiation opportunities, and prioritizing strategic brand positioning using large-scale consumer-generated data.

**Keywords:** Brand perception; Strategic brand positioning; Social media analytics; Latent Dirichlet allocation; Aspect-based sentiment analysis; Competitive intelligence

## 1 Introduction

Data-driven decision-making has become an important capability for firms operating in highly competitive markets. Organizations increasingly rely on digital data sources to understand customer perceptions, monitor competitors, and make strategic positioning decisions. Social media platforms have become one of the most important sources of external market intelligence because consumers continuously generate opinions, experiences, recommendations, and evaluations through online interactions [1, 2]. The scale of digital interaction continues to grow. According to recent reports, there were about 5.24 billion active social media user identities in the world in 2025 [3]. This figure increased by more than 4% compared with the previous year. Thus, social media is one of the largest repositories of consumer-generated market information that can be accessed by firms.

This development has led to a fundamental change in brand competition. Consumers do not rely on corporate communications alone. They actively create, share, and interpret brand-related content. User-generated content has a strong impact on brand image, brand awareness, brand trust, and purchase behavior. Previous studies show that social media communication and user-generated content strongly affect brand associations, customer engagement, and purchase intentions [2, 4, 5]. The beauty and personal care industry is a good example of this phenomenon. The industry is characterized by intense competition, fast innovation, short product life cycles, and high consumer engagement [1, 4]. Consumers discuss the effectiveness of skincare, ingredients, fragrances, texture, packaging, pricing, influencer recommendations, and brand campaigns on social media platforms [6]. These discussions create digital traces that reflect how consumers perceive and evaluate competing brands and products and provide valuable information about consumers' preferences and market dynamics [1, 4]. More recent evidence shows that beauty consumers use social media content, reviews, influencer communication, and peer recommendations to evaluate beauty brands and products [7–9]. The content is perceived as more credible and reliable than any kind of firm-generated information and, thus, plays an important role in shaping customer attitudes and purchase behavior.

Brand perception is a key element of strategic brand positioning. Customer-based brand equity theory emphasizes that the firms try to create favorable, strong, and unique brand associations in consumers' minds [10]. Strategic positioning is based on the way in which consumers perceive and differentiate competing brands. However, brand perception is inherently multidimensional. Consumers evaluate the quality of the product, its functionality, effectiveness, price, sensory experience, image, and symbolic value at once. Positive evaluation in one dimension does not mean positive evaluations in other dimensions.

Social media analytics became an effective method for collecting brand-related intelligence from digital conversations. Sentiment analysis is one of the most popular approaches because it allows classifying opinions into positive, neutral, and negative categories [11, 12]. Previous studies show that sentiment analysis can provide valuable information about customer response to brands, products, and services [13]. However, sentiment analysis provides only partial strategic information because it does not explain what consumers discuss or which features affect brand perception [14]. Topic modeling provides another perspective for brand perception analysis because it allows to discover the dominant themes within large text collections [15, 16]. Advances in natural language processing have led to the introduction of latent Dirichlet allocation-based topic modeling approaches, especially for short social media texts [17]. Aspect-based sentiment analysis provides another level of analysis that allows to distinguish features for which customers express positive or negative opinions. Aspect-based sentiment analysis becomes increasingly popular in e-commerce, service management, and social media analytics research [14, 18].

However, several important research gaps still exist. First, existing studies usually analyze brand perception mostly on the brand level, but a limited number of studies have treated brand perception as a competitive construct. Second, previous studies have analyzed sentiment, topics, or aspects separately. Limited studies have integrated them in order to create a unified framework for competitive brand perception analysis. Third, existing studies have rarely turned the social media analysis outputs into strategic positioning indicators for managerial decision-making. Fourth, limited research has described how digital conversations can reveal competitive differentiation, perception gaps, and strategic market opportunities across competing brands. Competitive brand positioning requires a deeper understanding of what consumers discuss, how they evaluate specific product attributes, and how these evaluations vary among competing brands.

In order to close the gap, this study proposes a data-driven competitive brand perception framework for strategic brand positioning. The framework integrates sentiment analysis latent Dirichlet allocation-based topic modeling, and aspect-based sentiment analysis and transforms unstructured social media conversations into actionable competitive intelligence. The framework evaluates competitive brand perception through three complementary competitive positioning indicators: topic salience, aspect sentiment, and brand distinctiveness. The combined interpretation of these indicators is then used to identify perception gaps and competitive whitespace as strategic insights for brand positioning. These indicators provide a multidimensional description of consumer perception of competing brands and of how it affects brand positioning. The framework is tested on X (formerly Twitter) conversations about four major beauty and personal care brands: Dove, Nivea, Estée Lauder, and L'Oréal. These brands occupy different market positions, have different product portfolios, and attract different consumer segments within the beauty and personal care market. Their large volume of digital conversations provides a good context for competitive brand perception analysis through social media conversations.

The study makes three contributions to the literature. First, it makes a contribution to brand management by conceptualizing brand perception as a competitive and multidimensional construct. Second, it makes a methodological contribution by integrating sentiment analysis, latent Dirichlet allocation-based topic modeling, and aspect-based sentiment analysis in one analytical framework. Third, it makes a managerial contribution by proposing a practical approach to reveal competitive strengths, perception gaps, and strategic positioning opportunities from social media data. The proposed competitive brand perception framework enables firms to go beyond sentiment analysis and develop an evidence-based brand positioning strategy.

## **2 Literature Review**

### **2.1 Brand Perception and Strategic Brand Positioning**

Brand perception is defined as consumers' evaluations, associations, beliefs, and experiences related to the brand. It affects consumers' differentiation of brands and their purchases. Customer-based brand equity theory states that favorable, strong, and unique brand associations create brand equity and competitive advantage [10]. In highly competitive markets, brand perception is also important in strategic brand positioning because companies try to occupy distinctive positions in customers' minds.

The development of digital platforms changes the way brand perceptions are formed and communicated. Consumers share their product experiences, recommendations, and evaluations through social media. These interactions generate large amounts of user-generated content reflecting consumer perceptions in real time. Recent studies show that social media communication strongly affects brand image, customer engagement, brand trust, and purchase intentions [4, 5]. Social media data becomes an important source of competitive market intelligence for brand management. Brand perception is inherently multidimensional. Consumers evaluate brands based on product quality, functionality, price, sensory experience, innovation, and symbolic value. Therefore, analysis of brand perception requires an approach that captures multiple dimensions of consumer evaluation instead of using overall brand sentiment.

### **2.2 Social Media Analytics and Consumer Intelligence**

Social media analytics is a set of computational methods for the extraction of meaningful information from digital conversations. The growing amount of user-generated content encourages organizations to use data-driven methods for understanding consumer behavior, monitoring competitors, and making strategic decisions. Social media analytics is widely applied in marketing, customer relationship management, innovation management, and competitive intelligence research. Previous studies show that social media data can be a valuable source of information about consumer preferences, brand interactions, and market trends [1, 2]. Unlike traditional surveys, social media analytics allows observing consumer opinions spontaneously generated in a natural environment [4, 16]. These digital traces allow researchers to understand consumers' opinions without interference from researchers or organizations.

Recent advances in natural language processing increase the capabilities of social media analytics. Methods such as sentiment analysis, topic modeling, and opinion mining help researchers to turn unstructured textual data into actionable market intelligence [15, 19, 20]. These methods become increasingly important for companies trying to understand dynamic consumer perceptions in highly competitive markets.

### **2.3 Topic Modeling for Brand Perception Mining**

Topic modeling is one of the most popular techniques for the discovery of latent themes in large text corpora. Traditionally, latent Dirichlet allocation, one of the most popular topic modeling approaches, is widely used in social media analytics and consumer research [13, 21–23]. Latent Dirichlet allocation identifies the groups of words that tend to co-occur in the text and represents them as latent topics. Even though latent Dirichlet allocation is widely used, it has some limitations when applied to short and noisy social media texts. Such texts lack co-occurrence information that reduces the interpretability of latent Dirichlet allocation topics. In order to overcome these limitations, recent studies have introduced latent Dirichlet allocation-based topic modeling approaches that leverage transformer-based language models and semantic embeddings. Another popular topic modeling approach is BERTopic, which uses transformer embeddings, dimensionality reduction, clustering algorithms, and class-based term frequency-inverse document frequency for generating semantically coherent topics [17].

For brand perception analysis, topic modeling helps to discover the dominant discussion themes associated with competing brands using latent Dirichlet allocation. This helps to identify the attributes, products, and experiences that affect consumer perceptions in digital environments.

### **2.4 Aspect-Based Sentiment Analysis**

Sentiment analysis is a popular method for the classification of opinions into positive, neutral, and negative categories. This technique is widely used for the evaluation of consumer responses to brands, products, and services [11, 15, 18, 24]. Even though sentiment analysis gives valuable information about overall consumer attitudes, it often cannot explain why consumers give positive or negative evaluations. Aspect-based sentiment analysis overcomes this limitation by assigning sentiment to individual aspects or attributes mentioned in the text. Instead of classifying the whole document into one of three sentiments (positive, negative, or neutral), aspect-based sentiment analysis analyzes sentiment towards specific product attributes such as quality, price, packaging, effectiveness, or service experience [14, 25].

Recent studies have pointed out the growing importance of aspect-based sentiment analysis for consumer intelligence and social media analytics because it allows more granular understanding of customer evaluations. Aspect-based sentiment analysis is widely used in e-commerce, hospitality, service management, and product review

analysis. Recent systematic reviews also show that transformer-based approaches significantly improve aspect extraction and sentiment classification in consumer-generated content analysis [14]. For beauty and personal care brands, aspect-based sentiment analysis is particularly relevant because consumers discuss multiple product attributes at once. Consumers may appreciate skincare effectiveness and criticize price or packaging. Thus, aspect-level analysis provides more information than sentiment analysis.

## 2.5 Competitive Brand Perception Framework

Even though previous studies successfully applied sentiment analysis, topic modeling, and aspect-based sentiment analysis to consumer research, these methods are often used independently. Sentiment analysis allows evaluating the overall emotional orientation of consumers, topic modeling discovers dominant discussion themes, while aspect-based sentiment analysis evaluates sentiment associated with specific attributes. However, each method provides only partial picture of consumer perception [14, 26]. Studies in social media analytics note that reliance on one analytical perspective overlooks important dimensions of consumer evaluations and market intelligence [1, 4]. Strategic brand positioning requires a broader perspective. Organizations need to understand what consumers discuss, how consumers evaluate specific attributes, and how these evaluations differ across competing brands. Brand perception is inherently multidimensional because consumers evaluate functional, emotional, symbolic, and experiential attributes at once [4, 10]. Competitive brand perception emerges from the interaction of discussion themes, sentiment evaluation, and attribute-level assessment.

Previous studies have noted the importance of the integration of thematic and evaluative dimensions in perception analysis. Similarly, recent advances in aspect-based sentiment analysis have pointed out the importance of linking sentiment with specific product and service attributes in order to provide more actionable managerial insights [14]. Based on these perspectives, this study proposes the competitive brand perception framework as an analytical framework for competitive brand evaluation. The framework integrates sentiment analysis latent Dirichlet allocation-based topic modeling, and aspect-based sentiment analysis and turns social media conversations into competitive intelligence. Integration of multiple analytical perspectives allows obtaining a more comprehensive understanding of consumer perception and competitive positioning than a single-method approach [1, 17].

The framework evaluates competitive brand perception through three analytical indicators: topic salience, aspect sentiment, and brand distinctiveness. Perception gaps and competitive whitespace are derived from the interaction among these indicators to support strategic brand interpretation. Aspect sentiment evaluates consumers' opinions towards specific product attributes. Brand distinctiveness measures how unique a topic is associated with a brand. Perception gaps help to find discrepancies between the importance of a topic and consumer evaluation. Competitive whitespace reveals market opportunities not yet covered by the current market conversations. These indicators provide a multidimensional picture of competitive brand positioning in digital markets and extend social media analytics research to the application for strategic brand positioning [1, 4].

## 2.6 Research Gap and Conceptual Positioning

Previous studies have shown the importance of social media analytics for the understanding of consumer perceptions and market behavior (Table A1). According to Table A1, prior studies have already shed light on social media engagement, brand equity, consumer behavior, sentiment, and topic structure. Nonetheless, a thorough examination of these studies shows certain fragmentation in the existing literature. In particular, there is no research among those compared here which encompasses the beauty and personal care context, competitive brand analysis, text mining, and strategic brand positioning all together.

First of all, prior research on the brands tends to consider mainly brand equity, brand management, social media engagement, or consumer interaction with regard to the particular brand. Previous studies have provided valuable insights into brand equity and social media-based brand management [10, 26–29]. Still, this body of knowledge lacks conceptualization of the brand perception as a relative construct which evolves through the comparisons with competing brands. Hence, the existing literature is poor in the information about how consumers differentiate between particular brand and its competitors based on topics, attributes, and sentiments discussed in social media.

Second, the field of beauty and personal care remains insufficiently researched. Out of all the works considered in Table A1, only one study touches upon the beauty-related context [4]. Nonetheless, this study mostly focuses on luxury brand engagement and does not propose any comprehensive framework for comparing multiple competing beauty and personal care brands. The majority of prior research is mostly devoted to luxury fashion, general social media marketing, consumer engagement, or methodologies. Such a focus represents a contextual gap considering the fact that beauty and personal care products provoke very unique consumer perceptions concerning the product efficiency, ingredients, sensory qualities, price, brand image, and sustainability.

Third, prior research has predominantly used analytical techniques associated with social media analysis separately. Methodological researches have offered and implemented latent Dirichlet allocation, BERTopic, sentiment analysis, and aspect-based sentiment analysis (see [14, 17, 21, 25, 30–32]). However, these works mostly concentrate on



the performance or implementation of analytical methods. Very limited number of research papers combined topic modeling, overall sentiment analysis, and aspect-based sentiment analysis in one single framework. Such a methodological limitation hampers the ability of existing approach to detect not only what consumers discuss but also how they perceive the brands and what particular aspects or features create positive or negative perceptions.

Fourth, the majority of prior studies stop at the output of the descriptive or diagnostic analysis. They define topics, sentiments, level of engagement, or brand equity; however, very few works use these outputs for generating the operational metrics for brand positioning. As seen in Table A1, strategic brand positioning and competitive intelligence are not considered together in prior research. Thus, the existing social media analytics research cannot provide sufficient managerial guidance to a company on how it should make its brand more distinctive, how to react to the negative perceptions, how to differentiate from other brands, or how to uncover unutilized market opportunities.

Fifth, competitive intelligence based on consumer-generated data is also underdeveloped. Very few prior research papers have considered cross-brand topic salience, relative sentiment performance, brand distinctiveness, perception gaps, and competitive whitespace together. Without such indicators, managers may understand the perception of their brand but cannot assess whether it means a true competitive advantage, common category attribute, relative weakness, or unclaimed market space.

To address the mentioned contextual, methodological, analytical, and managerial gaps, this study suggests a data-driven framework of competitive brand perception for beauty and personal care industry. The suggested framework incorporates latent Dirichlet allocation-based topic modeling, overall sentiment analysis, and aspect-based sentiment analysis for capturing the structure of the consumer perceptions regarding particular brands. Unlike the approaches examining particular brands, the proposed framework considers brand perception as competitive and multi-dimensional construct which develops through inter-brand comparisons.

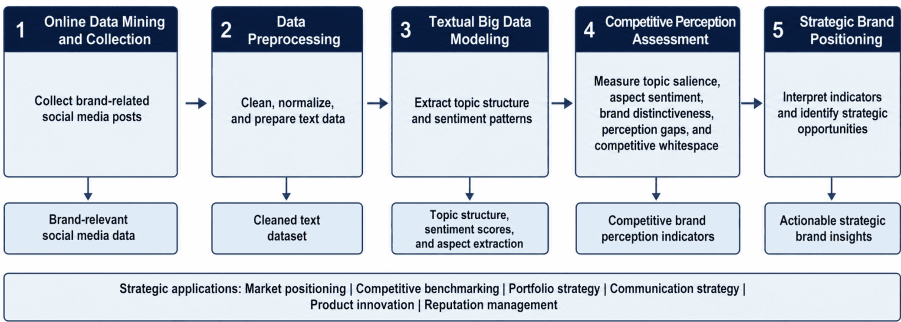
In addition, the proposed framework uses the outputs of social media analysis for generating the strategic indicators such as topic salience, aspect-level sentiment, brand distinctiveness, perception gaps, and competitive whitespace. This way, the indicators tie social media conversations with competitive intelligence and strategic decisions on brand positioning. Therefore, the novelty of this research does not only consist in combining multiple text-mining techniques but also in development of the end-to-end analytical framework which transforms unstructured conversations into competitive and managerially actionable brand-positioning insights.

As can be seen from Table A1, this study stands out among the compared works as the first one encompassing social media data, beauty and personal care context, competitive brand approach, topic modeling, sentiment analysis, aspect-based sentiment analysis, measurement of brand perception, strategic brand positioning, and competitive intelligence. This comprehensive position distinguishes the suggested framework from the prior research papers which covered only some of these dimensions.

### 3 Methodology

#### 3.1 Research Design

This study uses a computational social media analytics approach to develop and validate the competitive brand perception framework (Figure 1) for strategic brand positioning. The framework turns unstructured social media conversations into competitive brand intelligence using the integration of sentiment analysis, latent Dirichlet allocation-based topic modeling, and aspect-based sentiment analysis. The research design includes five stages. The first stage is data collection and preprocessing. The second stage is sentiment analysis and identification of overall consumer evaluations. The third stage is latent Dirichlet allocation-based topic modeling and identification of latent perception themes. The fourth stage is aspect-based sentiment analysis and evaluation of product and brand attributes. The fifth stage is synthesis of the analytical outputs into three competitive perception indicators—topic salience, aspect sentiment, and brand distinctiveness—which are subsequently interpreted to identify perception gaps and competitive whitespace.



**Figure 1.** Proposed competitive brand perception framework

### 3.2 Data Collection

The empirical context of this study is the beauty and personal care industry. Four global brands were selected: Dove, Nivea, Estée Lauder, and L'Oréal. These brands occupy different market positions, have different product portfolios, and attract different consumer segments within the beauty and personal care market. X (formerly Twitter) was chosen as a primary data source because it includes large volumes of publicly available user-generated content about consumer experiences, product evaluations, and brand interactions. The dataset included tweets collected during one year using brand-specific keywords.

The initial dataset included 152,848 tweets. Data preprocessing and relevance filtering were then performed in order to eliminate duplicates, irrelevant content, non-brand references, and marketplace noise. The final analytical corpus included 12,603 unique tweets.

### 3.3 Data Preprocessing

Social media texts can be noisy and include informal language, hyperlinks, hashtags, user mentions, emojis, and duplicated content. Therefore, several preprocessing steps were performed before analysis. The preprocessing procedure included:

- Lowercase conversion
- Uniform resource locator removal
- User mention removal
- Retweet marker removal
- Emoji and punctuation removal
- Whitespace normalization
- Duplicate tweet removal
- Brand relevance filtering

Brand relevance filtering was particularly important because there can be several meanings of a brand name outside the beauty category. For instance, Dove may be associated with a bird rather than with a personal care brand. Therefore, only tweets that included brand-related contextual cues were retained for analysis.

### 3.4 Sentiment Analysis

Sentiment analysis was performed in order to evaluate overall consumer attitude towards competing brands. Each tweet was assigned a sentiment score using a hybrid approach that combined lexicon-based sentiment scoring and transformer-assisted polarity classification. Tweets were classified into three sentiment categories:

- Positive
- Neutral
- Negative

### 3.5 Latent Dirichlet Allocation-Based Topic Modeling

Topic modeling was performed using latent Dirichlet allocation, one of the most popular probabilistic topic modeling techniques for the discovery of latent thematic structure in large text corpora [30]. Latent Dirichlet allocation was chosen because it allows to identify dominant discussion themes based on word co-occurrence patterns and has been widely applied in social media analytics, consumer intelligence, and brand perception research.

Even though recent studies have introduced latent Dirichlet allocation-based topic modeling approaches, such as BERTopic, latent Dirichlet allocation was considered as a more appropriate approach for this study because of its interpretability, wide usage in social media analytics research, and ability to produce interpretable topics suitable for competitive brand comparison. Additionally, the main goal of this study was not maximization of semantic representation performance, but identification of interpretable perception themes that could be used for strategic brand positioning.

The topic modeling procedure consisted of four stages:

- Document-term matrix construction
- Determination of the optimal number of topics
- Topic extraction using latent Dirichlet allocation
- Topic interpretation and representation

First, preprocessed tweets were turned into the document-term matrix using bag-of-words representation. The frequencies of terms were computed after preprocessing, which included tokenization, stopword removal, and lemmatization. Second, the optimal number of topics was selected based on topic coherence evaluation. Several topic models were evaluated and the model with the highest coherence score and interpretability was chosen for further analysis. In order to select the optimal number of topics, several models were evaluated with topic numbers ranging from 3 to 15 using topic coherence scores. Coherence was selected as a topic model evaluation metric because it measures semantic consistency of topic keywords and is widely used for this purpose. The model with the highest

coherence score and interpretability was retained for further analysis. Third, the latent Dirichlet allocation model was applied to compute latent topic distributions in the corpus. Latent Dirichlet allocation assumes that each document is represented as a mixture of topics and each topic is characterized by the distribution of words. Finally, discovered topics were interpreted based on their most probable keywords and representative tweets. Topic labels were assigned manually in order to improve interpretability and consistency with the beauty and personal care context.

The output of the latent Dirichlet allocation model included the following elements:

- Topic labels
- Topic frequencies
- Topic distributions
- Topic salience scores
- Representative keywords
- Representative tweets

The discovered topics were integrated with sentiment analysis and aspect-based sentiment analysis in order to develop the competitive brand perception framework. Thus, topic modeling within the competitive brand perception framework was implemented using a latent Dirichlet allocation-based approach during all analysis stages. Therefore, all references to topic modeling in this study refer to latent Dirichlet allocation-based topic modeling.

### 3.6 Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis was performed in order to evaluate sentiment towards specific product and brand attributes. Based on prior studies on aspect-based sentiment analysis and consumer perception mining [14, 33–35], a domain-specific aspect taxonomy was created using a literature review and exploratory analysis of the collected tweets. The aspect categories were designed in order to cover the main product and brand dimensions discussed in the beauty and personal care industry. The final taxonomy included the following categories:

- Body care and cleansing
- Facial skincare
- Hair care
- Makeup and beauty
- Sun protection
- Deodorant and freshness
- Brand image and campaign
- Price and promotion
- Men care

Tweets were assigned to one or more aspects using semantic similarity and keyword matching procedures. Aspect sentiment was then computed by aggregation of sentiment scores within each aspect category. The average sentiment score for aspect ( $\alpha$ ) is calculated as:

$$\bar{S}_a = \frac{1}{n_a} \sum_{i=1}^{n_a} S_i \quad (1)$$

where,  $\bar{S}_a$  is the average sentiment score of aspect  $a$ ;  $n_a$  is the number of tweets associated with aspect  $a$ ; and  $S_i$  is the sentiment score of tweets ( $i$ ).

The results can be used to evaluate overall consumer attitudes towards each brand.

### 3.7 Competitive Brand Perception Framework

The competitive brand perception framework integrates outputs from sentiment analysis, latent Dirichlet allocation, and aspect-based sentiment analysis. The framework evaluates competitive brand perception through three analytical indicators: topic salience, aspect sentiment, and brand distinctiveness. Perception gaps and competitive whitespace are subsequently derived as strategic insights from the combined interpretation of these indicators.

Topic salience measures the relative importance of a topic within a brand, which represents the proportion of tweets associated with topic ( $t$ ) relative to the total number of tweets associated with brand ( $b$ ).

$$TS_{bt} = \frac{N_{bt}}{N_b} \quad (2)$$

As a relative measure, the value of topic salience is in range of  $0 \leq TS_{bt} \leq 1$ . Higher value indicates that the topic dominates conversations about the brand and, therefore, reflects the dominant discussion theme among consumers. The metric can also be expressed as a percentage by multiplying the value of ( $TS_{bt}$ ) by 100. Aspect sentiment measures consumer evaluation toward a specific aspect and is represented by the average sentiment score of

tweets associated with that aspect. Brand distinctiveness measures how strongly a topic is associated with a brand compared with the overall market.

$$BD_{bt} = \frac{TS_{bt}}{TS_t} \quad (3)$$

The brand distinctiveness score is defined as the ratio between a topic's share within a brand and its overall share across the complete corpus, as shown in Eq. (3). Substituting Eq. (4) and Eq. (5) into Eq. (3) yields Eq. (6), which can be further simplified into Eq. (7). Eq. (8) and Eq. (9) define the computation of the total number of tweets assigned to each topic and the total corpus size, respectively. Distinctiveness ratios were calculated using the original (unrounded) topic proportions, while percentages presented in the table are rounded for readability.

$$TS_{bt} = \frac{N_{bt}}{N_b} \quad (4)$$

$$TS_t = \frac{N_t}{N} \quad (5)$$

$$BD_{bt} = \frac{\frac{N_{bt}}{N_b}}{\frac{N_t}{N}} \quad (6)$$

$$BD_{bt} = \frac{N_{bt}N}{N_bN_t} \quad (7)$$

$$N_t = \sum_{b=1}^B N_{bt} \quad (8)$$

$$N = \sum_{b=1}^B N_b \quad (9)$$

where,  $BD_{bt}$  is the brand distinctiveness score of topic  $t$  for brand  $b$ ;  $N_{bt}$  is the number of tweets assigned to topic  $t$  within brand  $b$ ;  $N_b$  is the total number of tweets belonging to brand  $b$ ;  $N_t$  is the total number of tweets assigned to topic  $t$  across all brands;  $N$  is the total number of tweets in the corpus; and  $B$  is the total number of brands.

$BD_{bt} > 1$  indicates that topic ( $t$ ) is overrepresented within brand ( $b$ ) relative to the overall market.  $BD_{bt} = 1$  indicates proportional representation, while  $BD_{bt} < 1$  indicates that topic ( $t$ ) is less associated with brand ( $b$ ) than with the overall market. Therefore, higher values of  $BD_{bt}$  indicate more competitive differentiation of a topic by a brand.

## 4 Results and Discussion

### 4.1 Descriptive Statistics and Sentiment Analysis

The final analytical corpus included 12,603 tweets after duplicates removal, brand relevance filtering, and noise reduction. Nivea generated the largest number of tweets (9,003), followed by Estée Lauder (1,696), Dove (1,471), and L'Oréal (433). There were initially 152,848 tweets collected through keyword-based X (formerly Twitter) scraping. After preprocessing, duplicates, irrelevant brand mentions, promotional or spam content, and noisy tweets with insufficient textual information were eliminated. Filtering process reduced the dataset from 152,848 tweets to 12,603 tweets suitable for analysis. A detailed summary of the filtering process is provided in Table 1 in order to increase transparency and reproducibility.

Examples of excluded content included tweets referring to the meaning of a keyword outside the brand context (e.g., "dove" as a bird reference), duplicated promotional messages, automated marketing tweets, and tweets mainly consisting of emojis, uniform resource locators, and noise. The significant decrease of the dataset size from 152,848 tweets emphasizes the importance of data cleaning and brand relevance filtering for the valid perception analysis. The filtering process summarized in Table 1 prioritized dataset quality over quantity by retaining only unique, brand-relevant consumer discussions.

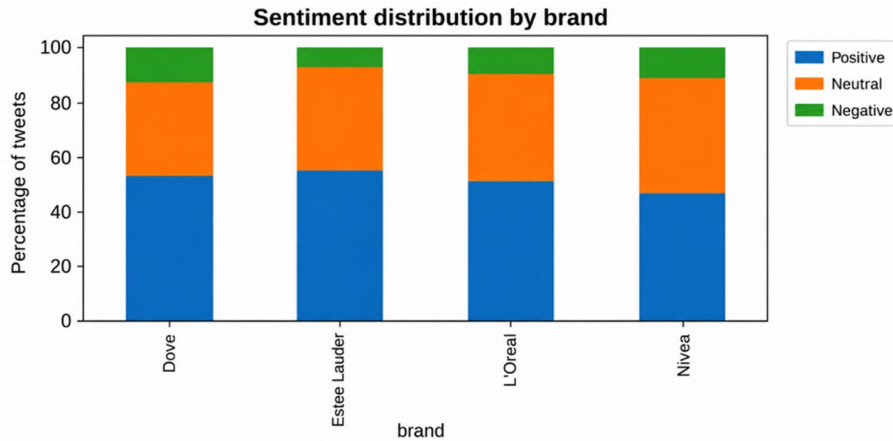


**Table 1.** Summary of filtering process

| Brand        | Raw Collected Tweets | Duplicate Tweets Removed | Brand-irrelevant Tweets Removed | Spam/Advertisements Removed | Noise & Short Text Removed | Final Analytical Corpus | Retention (%) |
|--------------|----------------------|--------------------------|---------------------------------|-----------------------------|----------------------------|-------------------------|---------------|
| Dove         | 65,253               | 35,904                   | 25,772                          | 1,482                       | 624                        | 1,471                   | 2.25          |
| Estée Lauder | 7,055                | 4,668                    | 479                             | 128                         | 84                         | 1,696                   | 24.04         |
| L'Oréal      | 1,545                | 437                      | 515                             | 97                          | 63                         | 433                     | 28.03         |
| Nivea        | 78,995               | 61,479                   | 7,635                           | 633                         | 245                        | 9,003                   | 11.40         |
| <b>Total</b> | <b>152,848</b>       | <b>102,488</b>           | <b>34,401</b>                   | <b>2,340</b>                | <b>1,016</b>               | <b>12,603</b>           | <b>8.25</b>   |

Although the retention rates varied across brands, the filtering procedure was intentionally designed to maximize data relevance rather than data quantity. Duplicate posts, advertisements, non-English or irrelevant content, and noisy texts were systematically removed. The particularly low retention rate for Dove mainly resulted from the frequent occurrence of the word “dove” in non-brand contexts (e.g., references to the bird or symbolic expressions), which substantially increased the number of brand-irrelevant posts removed. Similarly, Nivea exhibited a high proportion of duplicate posts, largely due to repeated promotional and reposted content. Consequently, the remaining corpus represents brand-relevant consumer-generated discussions suitable for subsequent analysis.

Figure 2 presents the distribution of sentiment for four brands of beauty and personal care items. On the whole, the sentiment towards all brands is predominantly positive and reflects positive consumer attitudes towards the products of particular companies. Thus, the share of positive sentiment constitutes about half of all brand-related tweets, while the negative sentiment is lower than 15% for all brands. It means that people tend to share their favorable experience and evaluate positively the products of beauty and personal care brands. Estée Lauder exhibited the highest proportion of positive sentiment (54.8%) and the lowest proportion of negative sentiment (7.1%) among the four brands, indicating the most favorable overall consumer evaluations. Such a result means that consumers perceive the brand as the source of positive product experience and high brand value. Thus, the sentiment distribution corresponds to the strategy of Estée Lauder, according to which product quality, brand reputation, and consumer satisfaction influence the positive perception of the brand by consumers.

**Figure 2.** Sentiment distribution by brand

The distribution of Dove and L'Oréal is rather similar (Table 2). Dove shares 53% of positive sentiment and L'Oréal — 50.8%. The brands have low shares of negative sentiment that means the stability of consumer attitudes. Nevertheless, Although Dove showed a slightly higher proportion of positive tweets (53.0%) than L'Oréal (50.8%), L'Oréal achieved a higher mean sentiment score (0.120 vs. 0.112), suggesting stronger overall sentiment intensity. Therefore, it can be concluded that both brands have created positive images, but different factors influence consumer perceptions of them. Nivea is characterized by the smallest share of positive sentiment and the largest share of neutral sentiment among four beauty brands. Thus, almost 47% of brand-related tweets are classified as positive, while 42.1% are classified as neutral. It means that consumers tend to talk about the products of Nivea without expressing any sentiments about them. This distribution of positive and negative sentiments is probably connected with the variety of Nivea products and its broad presence in the market.

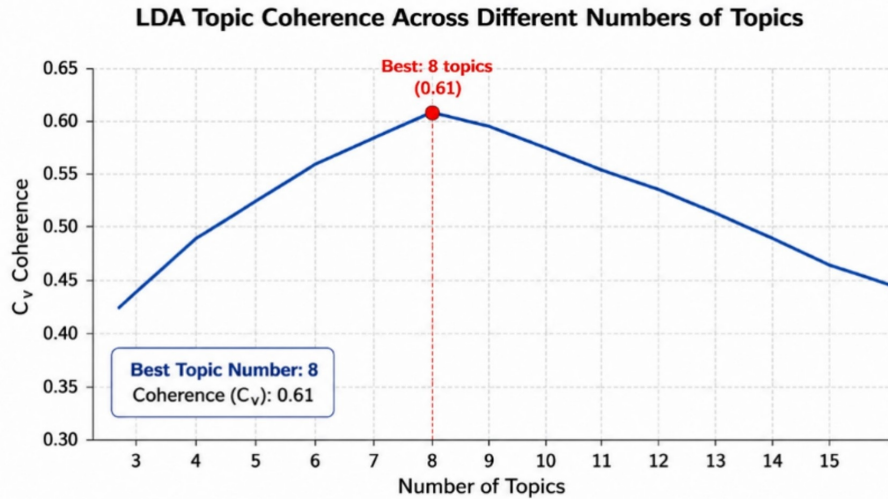
**Table 2.** Sentiment summary by brand

| Brand        | Tweets | Mean Sentiment | Positive (%) | Neutral (%) | Negative (%) | Total Engagement | Mean Engagement | Engagement-Weighted Sentiment |
|--------------|--------|----------------|--------------|-------------|--------------|------------------|-----------------|-------------------------------|
| Dove         | 1,471  | 0.112          | 53.0         | 34.4        | 12.6         | 6,617            | 4.50            | 0.111                         |
| Estée Lauder | 1,696  | 0.132          | 54.8         | 38.1        | 7.1          | 18,824           | 11.10           | 0.136                         |
| L'Oréal      | 433    | 0.120          | 50.8         | 39.5        | 9.7          | 2,214            | 5.11            | 0.112                         |
| Nivea        | 9,003  | 0.097          | 46.6         | 42.1        | 11.3         | 74,641           | 8.29            | 0.101                         |

Sentiment distribution is presented in Table 2. Estée Lauder achieved the highest mean sentiment score (0.132), followed by L'Oréal (0.120), Dove (0.112), and Nivea (0.097). Estée Lauder also exhibited the highest proportion of positive sentiment (54.8%) and the lowest proportion of negative sentiment (7.1%), indicating the most favorable overall consumer evaluations among the four brands. Although Dove showed a slightly higher share of positive sentiment (53.0%) than L'Oréal (50.8%), L'Oréal obtained a higher mean sentiment score (0.120 vs. 0.112), suggesting that its positive opinions were expressed with greater overall sentiment intensity. Although Nivea had the largest number of consumer discussions and high engagement, it also had a more heterogeneous sentiment profile compared to Estée Lauder and Dove. Thus, high social media exposure does not necessarily translate into more favorable consumer evaluations. Despite generating the largest number of discussions (9,003 tweets), Nivea recorded the lowest mean sentiment score (0.097) and the lowest proportion of positive sentiment (46.6%). In contrast, Estée Lauder achieved the highest mean sentiment score (0.132), the highest positive sentiment share (54.8%), and the lowest negative sentiment share (7.1%) despite a substantially smaller discussion volume. The obtained data proves the hypothesis about the inefficiency of measuring brand perception by only volume of mentions. Concerning strategic positioning, Estée Lauder seems to be perceived as a premium brand, while Nivea is positioned as a mass-market brand. It is impossible to detect this difference using only sentiment analysis; thus, it is necessary to include additional layers into the analysis process.

#### 4.2 Topic Number Selection

Several latent Dirichlet allocation model candidates were tested to decide on the best topic structure. The topic coherence was calculated for models that have a varying number of topics ranging from 3 to 15 topics. As seen in Figure 3, the topic coherence improved until eight topics, then decreased gradually for higher numbers of topics. Eight topics gave the best topic coherence and maintained the meaningful differences between topics at the same time.

**Figure 3.** Latent Dirichlet allocation topic coherence

Note: LDA = latent Dirichlet allocation; the highest coherence was obtained at  $K = 8$  ( $C_v = 0.61$ ), and the 8-topic model was therefore selected.

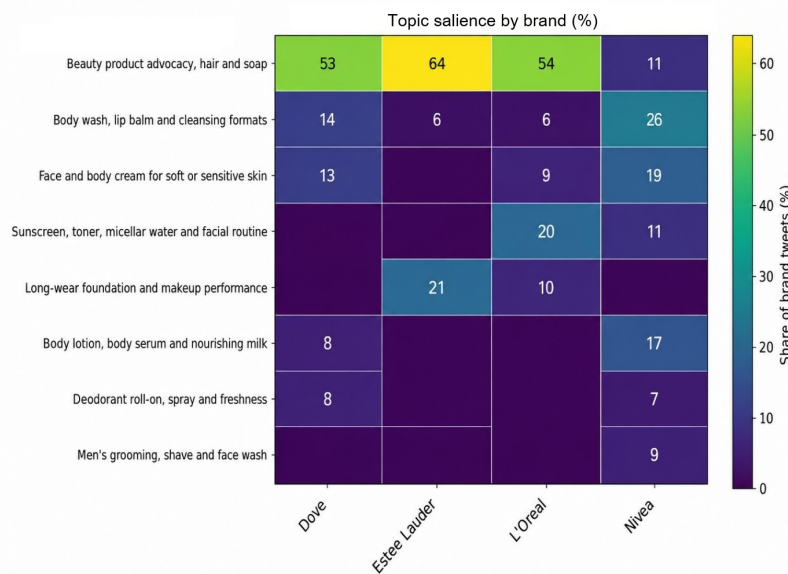
#### 4.3 Topic Structure of Competitive Brand Perception

The topic model based on latent Dirichlet allocation found eight meaningful topics that are able to cover the main ideas expressed in the discussion of beauty and skincare products. The largest topic is beauty product promotion, hair

& soap (24.7%), body wash, lip balm and cleaning products (20.8%), and face & body cream for soft/sensitive skin (15.7%) (Table 3 and Figure 4).

**Table 3.** Latent Dirichlet allocation-based topic model

| Topic | Label                                               | Tweets | Share (%) | Mean Sentiment |
|-------|-----------------------------------------------------|--------|-----------|----------------|
| 1     | Beauty product advocacy, hair and soap              | 3,109  | 24.7      | 0.127          |
| 4     | Body wash, lip balm and cleansing formats           | 2,624  | 20.8      | 0.083          |
| 5     | Face and body cream for soft or sensitive skin      | 1,980  | 15.7      | 0.113          |
| 2     | Body lotion, body serum and nourishing milk         | 1,718  | 13.6      | 0.106          |
| 0     | Sunscreen, toner, micellar water and facial routine | 1,091  | 8.7       | 0.067          |
| 3     | Men's grooming, shave and face wash                 | 886    | 7.0       | 0.086          |
| 7     | Deodorant roll-on, spray and freshness              | 746    | 5.9       | 0.116          |
| 6     | Long-wear foundation and makeup performance         | 449    | 3.6       | 0.141          |



**Figure 4.** Latent Dirichlet allocation-based topic salience by brand

Topics are presented in descending order of topic prevalence (number of tweets), whereas the topic numbers correspond to the original topic identifiers generated by the latent Dirichlet allocation model. The top keywords associated with each topic are provided in Table A2.

The first theme deals with promotion of beauty products such as hair products, soap, and discussion on other products. Another theme includes body wash, lip balm, and cleaning products. Other themes include face and body creams for sensitive skin, body lotion, and body serum, sunscreen and facial skincare, grooming products for men, deodorants that keep you fresh, and the effectiveness of long-wear foundation or makeup. The strongest sentiment can be seen in the effectiveness of long-wear foundation or makeup theme, as it coincides with very positive reviews on the durability of makeup.

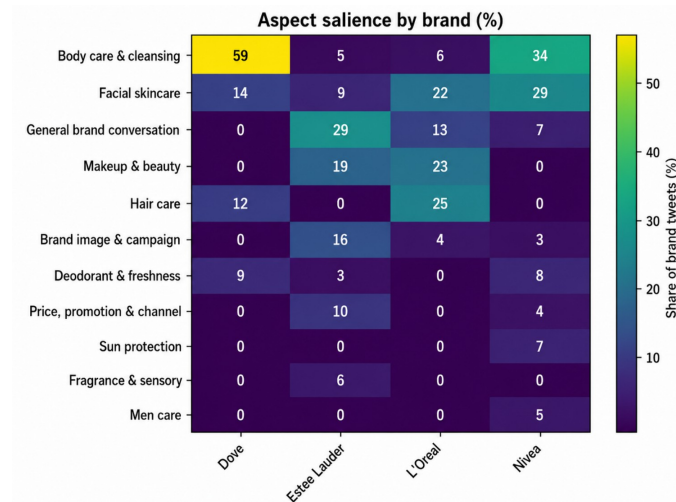
The topic hierarchy implies that consumer discussion is predominantly about products rather than the brand itself. Product-related topics such as product functionality, product types, and usage experience are discussed more often than company communication and brand identity concerns. The topic structure correlates with the utilitarian nature of beauty/personal care consumption. Long-wear foundation and makeup performance have the highest sentiment score (0.141), meaning that the consumers appreciate product functionality and performance. In contrast, sunscreen, toner, micellar water, and facial routine show the lowest sentiment score (0.067). This means that the competitive advantage depends not only on the presence of the topic but also on sentiment quality. Strong sentiment topics could mean the domains where the competitive advantages can be achieved sustainably.

#### 4.4 Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis provides more detailed insight into consumer evaluations by tying the sentiment to the specific product/brand aspects (Table 4).

**Table 4.** Top five aspects by brand

| Brand        | Aspect                     | Tweets | Share (%) | Mean Sentiment | Positive (%) | Negative (%) |
|--------------|----------------------------|--------|-----------|----------------|--------------|--------------|
| Dove         | Body care & cleansing      | 869    | 59.1      | 0.120          | 52.9         | 10.9         |
| Dove         | Facial skincare            | 202    | 13.7      | 0.094          | 55.0         | 16.3         |
| Dove         | Hair care                  | 180    | 12.2      | 0.070          | 42.8         | 17.8         |
| Dove         | Deodorant & freshness      | 137    | 9.3       | 0.122          | 56.9         | 13.9         |
| Dove         | Brand image & campaign     | 21     | 1.4       | 0.076          | 61.9         | 19.0         |
| Estée Lauder | General brand conversation | 495    | 29.2      | 0.131          | 53.9         | 5.7          |
| Estée Lauder | Makeup & beauty            | 329    | 19.4      | 0.098          | 48.0         | 6.4          |
| Estée Lauder | Brand image & campaign     | 263    | 15.5      | 0.106          | 49.8         | 8.7          |
| Estée Lauder | Price, promotion & channel | 168    | 9.9       | 0.122          | 58.3         | 9.5          |
| Estée Lauder | Facial skincare            | 153    | 9.0       | 0.165          | 66.0         | 7.2          |
| L'Oréal      | Hair care                  | 107    | 24.7      | 0.113          | 57.0         | 8.4          |
| L'Oréal      | Makeup & beauty            | 99     | 22.9      | 0.188          | 55.6         | 1.0          |
| L'Oréal      | Facial skincare            | 97     | 22.4      | 0.122          | 53.6         | 13.4         |
| L'Oréal      | General brand conversation | 55     | 12.7      | 0.022          | 25.5         | 21.8         |
| L'Oréal      | Body care & cleansing      | 25     | 5.8       | 0.087          | 52.0         | 20.0         |
| Nivea        | Body care & cleansing      | 3,092  | 34.3      | 0.096          | 44.4         | 10.3         |
| Nivea        | Facial skincare            | 2,609  | 29.0      | 0.096          | 50.6         | 15.6         |
| Nivea        | Deodorant & freshness      | 737    | 8.2       | 0.114          | 52.2         | 7.9          |
| Nivea        | Sun protection             | 645    | 7.2       | 0.093          | 40.2         | 9.6          |
| Nivea        | General brand conversation | 597    | 6.6       | 0.076          | 32.2         | 7.2          |

**Figure 5.** Aspect salience by brand

Aspect analysis points to different focal points for each brand category. Dove focuses on body care and cleansing, while its secondary aspects include facial skin care and hair care. Estée Lauder is primarily characterized by general brand conversation, followed by discussions related to makeup and beauty, brand image and campaign, price and promotion, and facial skincare; the latter includes a considerable amount of brand-related topics too. L'Oréal has a focus on hair care, makeup and beauty, as well as facial skin care. The broadest product coverage has been shown by Nivea.

Body care and cleansing comprised 59.1% of all aspect-specific consumer discussions related to Dove (Table 4 and Figure 5). This finding is consistent with Dove's established positioning based on personal care and hygiene of everyday items. The dominance of discussions regarding the aspect of cleansing is justified by the identity of the brand which implies gentle care, nourishing the skin and everyday healthiness. The research findings prove that the customers still associate Dove mainly with its main functional properties, thus there is a high congruency between the brand's intended positioning and consumers' perceptions of it. Estée Lauder demonstrated a distinctive perception profile. The largest share of consumer discussions was classified as general brand conversation (29.2%),

indicating that consumers frequently discussed the brand as a whole rather than focusing on a single product category. Makeup and beauty (19.4%) and brand image and campaign (15.5%) were the next most prominent aspects, reflecting consumers' attention to both the brand's cosmetic offerings and its marketing presence. Although facial skincare accounted for only 9.0% of the aspect-specific discussions, it achieved the highest mean sentiment score (0.165), suggesting particularly favorable consumer evaluations in this category. Overall, these findings are consistent with Estée Lauder's positioning as a premium beauty brand, where positive consumer perceptions extend beyond individual products to the brand's overall image and reputation.

L'Oréal showed equal distribution of consumer discussions between hair care, makeup and beauty, and facial skincare. However, the highest sentiment score (0.188) was observed for makeup and beauty which shows that cosmetic performance becomes one of the competitive advantages. Equal distribution between hair care, makeup, and facial skincare corresponds to the positioning of L'Oréal as a comprehensive beauty brand with diverse products. In contrast to those brands whose consumers' discussions were mostly concentrated in one aspect, L'Oréal managed to maintain a diverse perception structure. Nivea showed the broadest set of aspects. Body care and cleansing, facial skincare, deodorant, sun protection, and general brand discussions became the target of consumer discussions. This diversity of aspects reflects the mass-market positioning and diverse portfolio of the products of Nivea. Thus, the broad distribution of consumer discussions across several aspects confirms the mass-market positioning and big portfolio of Nivea. Unlike the brands with a dominant discussion of one aspect, the customers consider Nivea to be a versatile personal care brand addressing several aspects of skincare and hygiene. Thus, competitive differentiation occurs due to the aspect ownership. The customers associate the brands with different sets of product features, thus developing a certain perception structure for each brand within the same market.

#### 4.5 Competitive Brand Positioning and Distinctiveness Analysis

Topic distinctiveness analysis reveals how strongly a topic is associated with a specific brand relative to the overall market (Table 5).

**Table 5.** Leading topics and distinctiveness by brand

| Brand        | Topic                                               | Tweets | Brand Share (%) | Market Share (%) | Distinctiveness Ratio |
|--------------|-----------------------------------------------------|--------|-----------------|------------------|-----------------------|
| Dove         | Beauty product advocacy, hair and soap              | 786    | 53.4            | 24.7             | 2.17                  |
| Dove         | Body wash, lip balm and cleansing formats           | 200    | 13.6            | 20.8             | 0.653                 |
| Dove         | Face and body cream for soft or sensitive skin      | 185    | 12.6            | 15.7             | 0.801                 |
| Dove         | Body lotion, body serum and nourishing milk         | 125    | 8.5             | 13.6             | 0.623                 |
| Estée Lauder | Beauty product advocacy, hair and soap              | 1,078  | 63.6            | 24.7             | 2.58                  |
| Estée Lauder | Long-wear foundation and makeup performance         | 353    | 20.8            | 3.6              | 5.84                  |
| Estée Lauder | Body wash, lip balm and cleansing formats           | 101    | 6.0             | 20.8             | 0.286                 |
| Estée Lauder | Face and body cream for soft or sensitive skin      | 69     | 4.1             | 15.7             | 0.259                 |
| L'Oréal      | Beauty product advocacy, hair and soap              | 235    | 54.3            | 24.7             | 2.20                  |
| L'Oréal      | Sunscreen, toner, micellar water and facial routine | 86     | 19.9            | 8.7              | 2.29                  |
| L'Oréal      | Long-wear foundation and makeup performance         | 44     | 10.2            | 3.6              | 2.85                  |
| L'Oréal      | Face and body cream for soft or sensitive skin      | 37     | 8.5             | 15.7             | 0.544                 |
| Nivea        | Body wash, lip balm and cleansing formats           | 2,296  | 25.5            | 20.8             | 1.23                  |
| Nivea        | Face and body cream for soft or sensitive skin      | 1,689  | 18.8            | 15.7             | 1.19                  |
| Nivea        | Body lotion, body serum and nourishing milk         | 1,556  | 17.3            | 13.6             | 1.27                  |
| Nivea        | Beauty product advocacy, hair and soap              | 1,010  | 11.2            | 24.7             | 0.455                 |



Estée Lauder achieved the maximum distinctiveness ratio in long-wear foundation and makeup performance (5.84). Thus, this ratio shows that the topic is six times more discussed in relation to the brand than on the market as a whole, reflecting the high level of product differentiation and positioning as a premium cosmetic line. The L'Oréal brand displayed its distinctiveness in long-wear foundation and makeup performance (2.85) and in sunscreen, toner, micellar water, and facial routine (2.29). It reflects the brand positioning through its beauty products' efficacy and expert knowledge about skin care routines. The Dove brand has distinctiveness in beauty product advocacy, hair, and soap (2.17). It corresponds to the tradition of positioning this brand as one which sells personal care products and body cleansers. The Nivea brand displays distinctiveness in body wash, lip balm and cleansing formats (1.23), face and body cream for soft or sensitive skin (1.19), and body lotion and nourishing milk (1.27). In contrast to the previously described brands, Nivea's differentiation occurs in several topics, while Estée Lauder and Dove have only one topic with maximum distinctiveness. In general, this data implies that competitive positioning occurs via differentiating topics, not just the sentiments.

#### 4.6 Competitive Brand Perception Framework

The framework uses three analytical dimensions—topic salience, aspect sentiment, and brand distinctiveness—to evaluate brand competition. Perception gaps and competitive whitespace are then derived as strategic interpretation outputs from the interaction among these dimensions. The first one is topic salience which allows measuring the topics talked about by consumers the most often. The second dimension is aspect sentiment that assesses the evaluation of product aspects by consumers. Brand distinctiveness evaluates whether the topic is specifically linked with the brand under study. The next dimension is called perception gap. They appear when a highly salient topic is assessed by consumers quite weakly. Finally, competitive whitespace evaluates the perceptual opportunity in a highly valued area that is poorly owned by competitors. The empirical findings show that sentiment cannot explain competitive positioning. For example, Nivea produced the largest volume of conversation. However, it did not have the highest sentiment score. On the contrary, Estée Lauder produced a smaller number of conversations but had higher sentiment and distinctiveness scores. Hence, the findings confirm that competitive perception is a multidimensional construct consisting of visibility, evaluation, and distinction.

According to topic salience analysis, the conversations related to beauty and personal care are mostly product-driven rather than brand-oriented. The topics of beauty product advocacy, hair and soap make up 24.7% of all conversations. Other main topics are body wash, lip balm and cleansing formats, and face and body cream for soft or sensitive skin. These findings prove that consumers judge the brand based on their experience of product usage and performance. Aspect sentiment analysis provides deeper insights into the evaluation of particular product categories. Dove is strongly associated with body care and cleansing. Estée Lauder is characterized by stronger associations with makeup and beauty, brand image and campaign, and premium product discussions. L'Oréal is associated with hair care, makeup, and facial skincare. The aspect portfolio of Nivea is the widest and covers multiple personal care categories. Thus, the findings prove that brands occupy different perceptual space and compete based on different aspects.

The analysis of brand distinctiveness highlights the differences in competitive positioning. Estée Lauder shows the highest distinctiveness ratio in long-wear foundation and makeup performance (5.84). It means that the topics are nearly six times more common in Estée Lauder conversations than in general conversation. L'Oréal also has strong ownership of the topics such as makeup performance and facial care topics. Dove has distinctive positions in beauty product advocacy and hair and soap. In its turn, the distinctiveness of Nivea is spread across many topics that means a broad mass-market positioning of the brand. The framework also indicates the existence of perception gaps. For example, topics of sunscreen, toner, micellar water, and facial routine have a high volume of discussion but low sentiment scores compared to other topics. It means that consumers often talk about these products while expressing some concerns or dissatisfaction. Such topics are signs of problems that need to be solved by improving product quality or communication strategy.

Finally, the competitive whitespace approach implies the presence of opportunities in the highly valued but poorly owned perception areas. Topics that have high positive sentiment but low ownership of the dominant brand become strategic opportunities for differentiation. For instance, several skincare and facial care discussions have positive evaluations and are distributed across multiple brands. Therefore, companies can improve their competitive positioning through implementing the appropriate communication and innovation strategies. Thus, the competitive brand perception framework differs from conventional brand analytics because it turns social media conversations into competitive intelligence. While other sentiment measurement approaches consider just consumer evaluation, the framework measures consumer attention, evaluation, and distinction at the same time. As a result, managers get an opportunity to reveal perceptual strengths, weaknesses, gaps, and opportunities using consumer-generated big data.

#### 4.7 Discussion

A few more recently conducted studies support the value of applying a multi-faceted approach to analyzing consumer perceptions in digital environments. Social media analytics and sentiment analysis were increasingly

recognized as valuable tools that allow extracting useful insights from massive consumer conversations and making strategic decisions [35, 36]. These studies claim that sentiment data alone cannot explain the complexities of consumer perceptions because attitudes towards particular brands are often formed under the influence of several discussion themes and contexts. Research concerning Nivea and Estée Lauder coincides with recent findings regarding the non-equivalence of brand visibility and brand favorability. According to recent research in beauty and cosmetics markets, social media communication can improve brand awareness and purchase intention [37]. However, social media visibility does not necessarily translate into brand favorability, which depends more on product effectiveness, trustworthiness, and brand associations [38]. At the same time, it was found recently that the effect of social media communication on purchase intentions is mediated by positive brand associations rather than mere exposure.

The strong performance of Estée Lauder is consistent with the findings from the luxury and prestige beauty literature. Previous studies state that the perception of higher quality, credibility, and symbolism of the premium beauty brands allows creating more favorable consumer perceptions and increased brand loyalty [4, 26]. The prevalence of the themes and discussions related to the company's campaigns and influencers in the Estée Lauder conversations further confirms previous evidence about the role of social media communication, influencers, and stories in establishing the consumer-brand relationship and creating the brand equity [2, 5]. It means that the competitive advantage in the prestige beauty market is achieved not only by the functional product but also by the symbolic associations of the premium brand imagery [4, 27]. The emphasis on the topic-level analysis in this study is confirmed by recent findings based on advanced topic modeling techniques applied to the analysis of the social media conversations. Recent studies based on the application of topic modeling techniques, including latent Dirichlet allocation and newer embedding-based approaches, demonstrate that the thematic structure reveals important dimensions of the public perception that cannot be revealed by the analysis of sentiment scores alone [38]. In addition, recent research in fashion and beauty markets reveals that the combination of the sentiment and topic analysis helps to explain consumer preferences and trends better than either of the two techniques used separately.

Besides, the competitive brand perception framework developed in this study complements the existing literature on social media analytics because it integrates the sentiment, topics, and aspect evaluations for making strategic decisions about brand positioning. Recent studies emphasize the fact that organizations increasingly use social media intelligence for brand monitoring not only for reputation purposes but also for competitive positioning [39]. The present findings contribute to this emerging line of research by showing how topic ownership, aspect evaluations, and brand distinctiveness affect competitive brand perception in digital markets.

#### **4.8 Managerial Implications**

The findings have several managerial implications for brand managers operating in competitive consumer markets. First of all, the findings show that monitoring the overall sentiment alone is not enough for understanding competitive brand performance. Managers need to combine sentiment measures with the topic and aspect analysis in order to identify those themes and attributes that are responsible for the consumer perceptions. It is expected to move from brand reputation monitoring to strategic perception management.

Secondly, the findings suggest that competitive advantage is increasingly determined by the topic ownership rather than only the volume of the conversation. Estée Lauder had a lower volume of conversations than Nivea but demonstrated higher sentiment and topic distinctiveness. Therefore, managers need to focus not only on the increasing visibility but also on the reinforcement of topic ownership that is important for the desired brand positioning.

Third, the findings highlight the importance of aspect differentiation. Consumers relate Dove with cleansing and body care, Estée Lauder with premium beauty performance, L'Oréal with makeup and facial skincare, and Nivea with the wide range of the personal care category. Those associations are important strategic resources that should be used in communication, product development, and branding. Managers can use the aspect-level sentiment analysis in order to identify both strengths and weaknesses and take the necessary actions to strengthen their positions.

Fourth, the competitive brand perception framework provides managers with a tool for identifying competitive opportunities in advance. By combining topic salience with brand distinctiveness, managers can discover underutilized topics, changes in consumer interests, and whitespace opportunities for differentiation. Such capability is especially important in rapidly changing industries such as beauty and personal care where consumer preferences change quickly and competitive advantages are increasingly shaped by the conversations on digital platforms.

Finally, the framework serves as a tool for strategic positioning based on continuous social media monitoring, competitive perception benchmarking, and assessment of the consistency between consumer expectations and existing brand positioning. As a result, the competitive brand perception framework provides managers with a more flexible, data-driven, and evidence-based approach to managing brand perception.

#### **5 Limitations and Future Research**

This research should be understood within the following contextual limitations. Firstly, the analysis was performed based on X (formerly Twitter) conversations only, which is only one type of the social media environment. The

discussions of consumers on the other platforms such as Instagram, TikTok, Reddit, and YouTube may show different patterns of interactions and perception structure. Secondly, the empirical application of the framework was done in the field of beauty and personal care only. In other industries, consumers' perceptions can be determined by different product characteristics, decision-making processes, and competitive dynamics; therefore, the topic structures, aspect associations, and positioning patterns should not be generalized for other contexts. Thirdly, the conversations on social media can be different depending on the geographical location, culture, language group. Therefore, the competitive perception patterns revealed in this research can be different when used for another country and population. Future research can expand the competitive brand perception framework for different industries, geographic contexts, and social media platforms in order to test broader applicability and validity.

## 6 Conclusion

In this study, the competitive brand perception framework was developed and tested empirically for the strategic brand positioning using the social media data in the beauty and personal care industry. Using the integration of sentiment analysis, latent Dirichlet allocation-based topic modeling, and aspect-level sentiment analysis, the framework creates a multidimensional view of the consumer perceptions of competing brands in a digital environment. The findings show that brand perception is much broader than the overall sentiment evaluation. While all four brands had predominantly positive sentiment, the important differences in the topic ownership, aspect associations, and competitive distinctiveness appeared. Estée Lauder had the strongest positive perception and the highest topic distinctiveness that corresponded to its premium positioning. Dove was associated with cleansing and body care, which reflected its everyday-care functionality. L'Oréal was strong in the facial skincare and makeup performance, and Nivea had the broadest category coverage among all personal care subcategories. These findings demonstrate that brands compete with each other by means of unique combinations of product attributes, consumer experience, and symbolism rather than by sentiment scores.

The study makes the following contributions to the literature. Firstly, it considers the concept of brand perception as a competitive and multidimensional construct rather than a standalone brand-level phenomenon. Secondly, it develops a framework that incorporates sentiment analysis, topic modeling, and aspect-based sentiment analysis in the context of competitive evaluation. Thirdly, it introduces topic salience, aspect sentiment, and brand distinctiveness as the additional indicators of competitive brand positioning based on the social media conversations. The findings also show that the consumer-generated content can be converted into the competitive intelligence for strategic positioning. Therefore, the competitive brand perception framework can serve as a practical and scalable tool for monitoring the market perceptions, discovering differentiation opportunities, and making data-driven brand strategy. The future research can extend the framework by incorporating different social media platforms, longitudinal analysis, and advanced large language models.

## Data Availability

The data used to support the research findings are available from the corresponding author upon request.

## Conflicts of Interest

The author declares no conflicts of interest.

## References

- [1] Z. Xue, Q. Li, and X. Zeng, "Social media user behavior analysis applied to the fashion and apparel industry in the big data era," *J. Retail. Consum. Serv.*, vol. 72, p. 103299, 2023. <https://doi.org/10.1016/j.jretconser.2023.103299>
- [2] D. Cao, M. Meadows, D. Wong, and S. Xia, "Understanding consumers' social media engagement behaviour: An examination of the moderation effect of social media context," *J. Bus. Res.*, vol. 122, pp. 835–846, 2021. <https://doi.org/10.1016/j.jbusres.2020.06.025>
- [3] DataReportal, "Digital 2025: Global overview report," 2025. <https://datareportal.com/reports/digital-2025-global-overview-report>
- [4] X. Liu, H. Shin, and A. C. Burns, "Examining the impact of luxury brand's social media marketing on customer engagement: Using big data analytics and natural language processing," *J. Bus. Res.*, vol. 125, pp. 815–826, 2021. <https://doi.org/10.1016/j.jbusres.2019.04.042>
- [5] H. Shahbaznezhad, R. Dolan, and M. Rashidirad, "The role of social media content format and platform in users' engagement behavior," *J. Interact. Mark.*, vol. 53, no. 1, pp. 47–65, 2021. <https://doi.org/10.1016/j.intmar.2020.05.001>
- [6] A. L. Lestari and A. Hananto, "How do firms use social media: Topic modeling of Twitter brand posts of four Indonesian skincare brands," *ASEAN Mark. J.*, vol. 15, no. 2, pp. 14–42, 2023. <https://doi.org/10.7454/amj.v15i2.1168>

- [7] E. Djafarova and T. Bowes, “‘Instagram made Me buy it’: Generation Z impulse purchases in the fashion industry,” *J. Retail. Consum. Serv.*, vol. 59, p. 102345, 2021. <https://doi.org/10.1016/j.jretconser.2020.102345>
- [8] X. Y. Leung, J. Sun, and B. Bai, “Thematic framework of social media research: State of the art,” *Tour. Rev.*, vol. 74, no. 3, pp. 517–531, 2019. <https://doi.org/10.1108/TR-05-2018-0058>
- [9] C. Lou and S. Yuan, “Influencer marketing: How message value and credibility affect consumer trust of branded content on social media,” *J. Interact. Advert.*, vol. 19, no. 1, pp. 58–73, 2019. <https://doi.org/10.1080/15252019.2018.1533501>
- [10] K. L. Keller, “Conceptualizing, measuring, and managing customer-based brand equity,” *J. Mark.*, vol. 57, no. 1, pp. 1–22, 1993. <https://doi.org/10.1177/002224299305700101>
- [11] J. Guerreiro and P. Rita, “How to predict explicit recommendations in online reviews using text mining and sentiment analysis,” *J. Hosp. Tour. Manag.*, vol. 43, pp. 269–272, 2020. <https://doi.org/10.1016/j.jhtm.2019.07.001>
- [12] S. Li and Y. Li, “A sentiment analysis of online reviews based on the word alignment model: A product improvement perspective,” in *Proceedings of the 2018 2nd IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference*, Xi’an, China, 2018, pp. 2226–2231. <https://doi.org/10.1109/IMCEC.2018.8469724>
- [13] D. A. Purnama, Subagyo, and N. A. Masrurroh, “Online data-driven concurrent product-process-supply chain design in the early stage of new product development,” *J. Open Innov. Technol. Mark. Complex.*, vol. 9, no. 3, p. 100093, 2023. <https://doi.org/10.1016/j.joitmc.2023.100093>
- [14] G. D’Aniello, M. Gaeta, and I. La Rocca, “KnowMIS-ABSA: An overview and a reference model for applications of sentiment analysis and aspect-based sentiment analysis,” *Artif. Intell. Rev.*, vol. 55, no. 7, pp. 5543–5574, 2022. <https://doi.org/10.1007/s10462-021-10134-9>
- [15] B. Jeong, J. Yoon, and J.-M. Lee, “Social media mining for product planning: A product opportunity mining approach based on topic modeling and sentiment analysis,” *Int. J. Inf. Manag.*, vol. 48, pp. 280–290, 2019. <https://doi.org/10.1016/j.ijinfomgt.2017.09.009>
- [16] Subagyo, D. A. Purnama, N. A. Masrurroh, and R. R. Pratama, “Modeling dynamic consumer preferences in product attributes for social media-based product improvement planning,” *Malays. J. Consum. Fam. Econ.*, vol. 32, no. 1, pp. 104–140, 2024. <https://doi.org/10.60016/majcafe.v32.05>
- [17] M. Grootendorst, “BERTopic: Neural topic modeling with a class-based TF-IDF procedure,” *arXiv*, vol. abs/2203.05794, 2022. <https://doi.org/10.48550/arXiv.2203.05794>
- [18] S. A. Nugroho and S. Widiyanto, “Exploring electric vehicle adoption in Indonesia using zero-shot aspect-based sentiment analysis,” *Sustain. Oper. Comput.*, vol. 5, pp. 191–205, 2024. <https://doi.org/10.1016/j.susoc.2024.08.002>
- [19] H. P. Suresha and K. K. Tiwari, “Topic modeling and sentiment analysis of electric vehicles of Twitter data,” *Asian J. Res. Comput. Sci.*, vol. 12, no. 2, pp. 13–29, 2021. <https://doi.org/10.9734/ajrcos/2021/v12i230278>
- [20] R. Arifin and D. A. Purnama, “Identifying customer preferences on two competitive startup products: An analysis of sentiment expressions and text mining from Twitter data,” *J. Infotel*, vol. 15, no. 1, pp. 66–74, 2023.
- [21] H. Jelodar, Y. Wang, C. Yuan, X. Feng, X. Jiang, Y. Li, and L. Zhao, “Latent Dirichlet allocation (LDA) and topic modeling: Models, applications, a survey,” *Multimed. Tools Appl.*, vol. 78, no. 11, pp. 15 169–15 211, 2019. <https://doi.org/10.1007/s11042-018-6894-4>
- [22] A. S. Huzaifah, R. Nurhasanah, and R. F. Adriansyah, “Topic modelling on beauty product reviews using latent Dirichlet allocation,” *J. Ilmu Komput. Agri-Informatika*, vol. 12, no. 1, pp. 119–131, 2025.
- [23] B. W. Hartanto and I. B. Dharma, “Unsupervised topic labeling and opportunity model of social media data for enhancing automotive product design processes,” *Data Inf. Manag.*, p. 100103, 2025. <https://doi.org/10.1016/j.dim.2025.100103>
- [24] C. Grimalt-Álvaro and M. Usart, “Sentiment analysis for formative assessment in higher education: A systematic literature review,” *J. Comput. High. Educ.*, vol. 36, no. 3, pp. 647–682, 2024. <https://doi.org/10.1007/s12528-023-09370-5>
- [25] M. Wankhade, A. C. S. Rao, and C. Kulkarni, “A survey on sentiment analysis methods, applications, and challenges,” *Artif. Intell. Rev.*, vol. 55, no. 7, pp. 5731–5780, 2022. <https://doi.org/10.1007/s10462-022-10144-1>
- [26] A. J. Kim and E. Ko, “Do social media marketing activities enhance customer equity? An empirical study of luxury fashion brand,” *J. Bus. Res.*, vol. 65, no. 10, pp. 1480–1486, 2012. <https://doi.org/10.1016/j.jbusres.2011.10.014>
- [27] B. Godey, A. Manthiou, D. Pederzoli, J. Rokka, G. Aiello, R. Donvito, and R. Singh, “Social media marketing efforts of luxury brands: Influence on brand equity and consumer behavior,” *J. Bus. Res.*, vol. 69, no. 12, pp. 5833–5841, 2016. <https://doi.org/10.1016/j.jbusres.2016.04.181>
- [28] J. H. Kietzmann, K. Hermkens, I. P. McCarthy, and B. S. Silvestre, “Social media? Get serious! Understanding

the functional building blocks of social media,” *Bus. Horiz.*, vol. 54, no. 3, pp. 241–251, 2011. <https://doi.org/10.1016/j.bushor.2011.01.005>

- [29] A. M. Kaplan and M. Haenlein, “Users of the world, unite! The challenges and opportunities of social media,” *Bus. Horiz.*, vol. 53, no. 1, pp. 59–68, 2010. <https://doi.org/10.1016/j.bushor.2009.09.003>
- [30] D. M. Blei, A. Y. Ng, and M. I. Jordan, “Latent Dirichlet allocation,” *J. Mach. Learn. Res.*, vol. 3, pp. 993–1022, 2003. <https://www.jmlr.org/papers/v3/blei03a.html>
- [31] H. Zhang, H. Rao, and J. Feng, “Product innovation based on online review data mining: A case study of Huawei phones,” *Electron. Commer. Res.*, vol. 18, no. 1, pp. 3–22, 2018. <https://doi.org/10.1007/s10660-017-9279-2>
- [32] C. J. Hutto and E. Gilbert, “VADER: A parsimonious rule-based model for sentiment analysis of social media text,” in *Proceedings of the International AAAI Conference on Web and Social Media*, Ann Arbor, MI, USA, 2014, pp. 216–225. <https://ojs.aaai.org/index.php/ICWSM/article/view/14550>
- [33] Suhariyanto, R. Sarno, C. Fatichah, and R. Abdullah, “Aspect-based sentiment analysis: Natural language understanding for implicit review,” *Int. J. Electr. Comput. Eng.*, vol. 14, no. 6, pp. 6711–6722, 2024. <https://doi.org/10.11591/ijece.v14i6.pp6711-6722>
- [34] Y. C. Hua, P. Denny, J. Wicker, and K. Taskova, “A systematic review of aspect-based sentiment analysis: Domains, methods, and trends,” *Artif. Intell. Rev.*, vol. 57, no. 11, p. 296, 2024. <https://doi.org/10.1007/s10462-024-10906-z>
- [35] R. H. Chowdhury, “Sentiment analysis and social media analytics in brand management: Techniques, trends, and implications,” *World J. Adv. Res. Rev.*, vol. 23, no. 2, pp. 287–296, 2024.
- [36] Y. Nurfauzi and W. Wulandari, “Customer satisfaction on social media: Analysing sentiment and brand perception through big data,” *Int. J. Econ. Lit.*, vol. 3, no. 4, pp. 222–234, 2025. <https://sociohum.net/index.php/INJOLE/article/view/17>
- [37] D. O. Syalsabilla and S. Budiono, “Impact of marketing communication strategy through social media to increase brand awareness and purchase intention of beauty products,” *Enrich. J. Manag.*, vol. 14, no. 5, pp. 947–955, 2024.
- [38] Y. Qian, Y. Jiang, J. Shang, Y. Chai, and Y. Liu, “Why some products compete and others don’t: A competitive attribution model from the customer perspective,” *Decis. Support Syst.*, vol. 169, p. 113956, 2023. <https://doi.org/10.1016/j.dss.2023.113956>
- [39] P. Brooklyn, A. Olukemi, and C. Bell, “Social media sentiment analysis for brand reputation management,” 2024, SSRN Electronic Journal. <https://doi.org/10.2139/ssrn.4906218>

## Appendix

**Table A1.** State-of-the-art and novelty positioning

| Ref.              | SMD | BPC | CBC | TM | SA | ABSA | BP | SBP | CI | Research Object                                                            |
|-------------------|-----|-----|-----|----|----|------|----|-----|----|----------------------------------------------------------------------------|
| [10]              |     |     |     |    |    |      | ✓  | ✓   |    | Customer-based brand equity                                                |
| [26]              | ✓   |     |     |    |    |      | ✓  | ✓   |    | Luxury fashion brands                                                      |
| [27]              | ✓   |     |     |    |    |      | ✓  | ✓   |    | Luxury brand equity and consumer behavior                                  |
| [2]               | ✓   |     |     |    |    |      | ✓  |     |    | Social media engagement                                                    |
| [5]               | ✓   |     |     |    |    |      |    |     |    | Social media content and engagement                                        |
| [4]               | ✓   | ✓   |     | ✓  | ✓  |      | ✓  |     | ✓  | Luxury brand engagement using social media analytics                       |
| [1]               | ✓   |     |     | ✓  | ✓  |      |    |     | ✓  | Fashion consumer behavior in the big data era                              |
| [25]              |     |     |     |    | ✓  |      |    |     |    | Sentiment analysis methods and applications                                |
| [14]              |     |     |     |    | ✓  | ✓    |    |     |    | Aspect-based sentiment analysis reference model                            |
| [30]              |     |     |     | ✓  |    |      |    |     |    | Latent Dirichlet allocation topic modeling method                          |
| [21]              | ✓   |     |     | ✓  |    |      |    |     |    | Topic modeling applications                                                |
| [17]              | ✓   |     |     | ✓  |    |      |    |     |    | BERTopic topic modeling method                                             |
| [31]              |     |     |     |    | ✓  |      |    |     |    | Deep learning for sentiment analysis                                       |
| [32]              | ✓   |     |     |    | ✓  |      |    |     |    | Social media sentiment analysis                                            |
| [28]              | ✓   |     |     |    |    |      | ✓  | ✓   |    | Social media strategy and brand management                                 |
| [29]              | ✓   |     |     |    |    |      | ✓  | ✓   |    | Social media marketing and consumer interaction                            |
| <b>This study</b> | ✓   | ✓   | ✓   | ✓  | ✓  | ✓    | ✓  | ✓   | ✓  | Competitive brand perception framework for beauty and personal care brands |

Note: SMD = social media data; BPC = beauty/personal care context; CBC = competitive brand context; TM = topic modeling; SA = sentiment analysis; ABSA = aspect-based sentiment analysis; BP = brand perception; SBP = strategic brand positioning; CI = competitive intelligence.



**Table A2.** Latent Dirichlet allocation top keywords

| Topic | Top Keywords                                                                                                                                                  |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Soap, hair, beauty, products, love, shampoo, skincare, bar, good, tom, ford, companies, it's, tom ford, video                                                 |
| 4     | Lip, wash, balm, lip balm, care, body wash, body, pack, lipbalm, face, creme, face wash, shower, sun, wash pack                                               |
| 5     | Cream, skin, face, soft, moisturizer, body cream, body, dry, light, sensitive, hand, wash, face cream, soap, sensitive skin                                   |
| 2     | Lotion, body, body lotion, skin, oil, lotion body, body serum, extra, dry, body milk, white, serum, nourishing, milk, dry skin                                |
| 0     | Sunscreen, toner, moist, serum, skin, water, wardah, cleanser, somethinc, micellar, azarine, pink, oily, garnier, micellar water                              |
| 3     | Men, gel, shave, men sensitive, smelling, pack, sensitive, wash, face, smelling men, body, deep, men care, shave balm, body wash                              |
| 7     | Deodorant, roll, deodorant roll, women, fresh, pack, women deodorant, spray, deo, roll men, milliliters, men, pearl beauty, pearl, fresh active               |
| 6     | Success, worked, success worked, dreamed, dreamed success, wear, foundation, double, double wear, infallible, wear foundation, makeup, place, stay, wear stay |